

Global Mobile Reviews

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Abstract

With the mass adoption of mobile phones, countless consumer-generated reviews have been posted on various e-commerce websites. This results in huge amount of free data about product quality, hardware specifications and customer experience. Performing sentiment analysis on the review texts provides a scalable and low-cost way for vendors and manufacturers to measure customer satisfaction in close-to real time over traditional questionnaires. While most studies of mobile reviews are carried out on relatively small scale datasets, use binary sentiment classification tasks and either do not exploit, or exploit very sparsely the review text with product specification attributes as joint features. We address this deficiency by benchmark testing five supervised classification methods on five class star-rating prediction using over 50,000 recent mobile phone reviews and product specification attributes. These five methods include Naive Bayes, Logistic Regression, Support Vector Machine, Random Forest, and a fine-tuned BERT transformer model. We employ class-weighted loss function to mitigate the rating distribution imbalance. Our fine-tuned BERT obtains accuracy 91.4 percent and macro F1-score 0.903, which surpasses the best traditional machine learning method by 7.2 percent. The SHAP value analysis on our fine-tuned BERT demonstrates that language used to describe battery and camera performance is of particular importance to the extremity of consumer sentiment. Our paper provides a robust benchmark for this particular domain specific classification problem, and quantitative evidence about the most important factors impacting consumer sentiments of mobile phones.

Index Terms

sentiment analysis, mobile device reviews, BERT transformer, natural language processing, machine learning classification

I. INTRODUCTION

Over the past two decades, the consumer electronics industry is undoubtedly one of the most competitive and rapid changing markets in the world owing to the tremendous proliferation in the penetration of smart phone and mobile devices. Since there are numerous mobile phone manufacturers releasing various models of cell phones at regular intervals with minute variations in their hardware attributes and incorporation of novel software features, buyers heavily rely on the comments/reviews posted in e-commerce websites, tech blogs/communities and social media networks in making their purchasing decision. Each of these sites contain millions of user reviews for various models of mobile phones (e.g. Thousands of reviews on Amazon, Flipkart, GSMArena), which is an untapped treasure for mining information regarding user satisfaction, features' preference and product quality. Natural Language Processing and its sub-field i.e. Sentiment Analysis can be used to provide a systematic and large-scale solution for analyzing the free-text reviews and deriving useful insights to various stakeholders such as product managers and marketing analysts without human intervention.

The relevance of automated sentiment analysis to the area of mobile devices today cannot be overemphasized. With worldwide smartphone shipments at 1.2 billion, and typical product lifetimes shrinking to as little as 12 to 18 months, manufacturers and retailers can potentially gain significant competitive advantages by having the ability to rapidly and exhaustively study and react to consumer sentiment. Unlike time-consuming and expensive surveys to measure customer satisfaction, automated analysis of reviews can be used to track sentiment for literally thousands of products and millions of reviews concurrently, in near real-time. In addition to sentiment polarity, recent review datasets also contain specification metadata that can allow for a more fine-grained analysis, relating the change in sentiment to a particular specification (e.g. Battery, camera, processor, software UI, etc.), which could not otherwise be performed.

A. Related Work

The concept of sentiment analysis for product reviews has been being researched in the natural language processing field since 2000. As with many areas, a small number of initial papers set the scene for many subsequent contributions in three fields of polarity detection through lexicons, opinion mining based on features and machine learning for sentiment classification: Pang

and Lee(2002) proved the viability of supervised machine learning classification through the use of bag-of-words models; they used support vector machines to perform a binary classification of movie reviews and reported results of classification accuracy far higher than previous rule-based methods. Work proceeded in extending these principles to product reviews and Hu and Liu(2004) pioneered several of the techniques used in feature-based opinion mining to provide information on aspects of products such as screen, battery and design. For classification specifically of mobile phone reviews, in using data from various sites such as twitter, amazon and specific technology sites, the earlier research only had access to methods that were 75-85 percent accurate and that required manual labeling of thousands of product reviews to be usable. With the recent advances in deep learning and methods using recurrent neural networks (e.g. LSTMs, GRUs) which can process sequences of words more efficiently and extract richer contextual information than previous approaches, accuracy increased by 5-10 percent over standard benchmarks compared to existing baselines. The introduction of large, pre-trained transformer-based models (e.g. BERT, RoBERTa, DistilBERT) pushed this ceiling even further with many achieving accuracy levels well above 90 percent.

A summary of this relevant work is illustrated in Table I.

TABLE I
LITERATURE REVIEW TABLE SHOWING THE CONTRIBUTIONS OF VARIOUS AUTHORS FOR SENTIMENT ANALYSIS OF MOBILE DEVICE REVIEWS.

Paper Name	Dataset Used	Method	Accuracy	Sentiment Classes	Spec. Features	Limitation
Pang & Lee [1]	Movie Reviews	SVM, NB	82.9%	Binary	×	Domain-specific; small corpus
Hu & Liu [2]	Amazon Products	Rule-based	N/A	Aspect-level	×	No statistical model
Yadav et al. [3]	Twitter Mobile	LSTM	84.3%	3-class	×	Short text only
Zhang et al. [4]	Amazon Phone Reviews	BERT	89.1%	5-class	×	English only
Singh & Sharma [5]	Flipkart Reviews	RF + TF-IDF	81.5%	Binary	×	Outdated dataset
Kumar et al. [6]	GSMarena Reviews	CNN-LSTM	86.7%	3-class	×	Spec features ignored
Proposed Study	Kaggle Mobile Reviews 2025	BERT + ML Ensemble	91.4%	5-class	✓	—

B. Dataset

For this project we used "Mobile Reviews: Sentiment and Specification Dataset (2025 Edition)" from Mohan Krishna Thalla (Kaggle) available at <https://www.kaggle.com/datasets/mohankrishnathalla/mobile-reviews-sentiment-and-specification> [14]. This dataset contained about 50,000 consumer mobile reviews where for each review there are approximately 25 feature columns regarding both free text of the review and structured product specification meta-data. Among textual features there is a free text review column, review summary and purchased with a verified status, among the specification features there are fields for brand name, model name, operating system version, RAM, internal storage, battery, camera megapixels, display size and processor type. In our classification problem we decided to use the star rating (from 1 to 5) as our label, so this is an ordinal 5-class classification problem. In our case data from the above mentioned dataset is representative of the current market of mobile phones and contains many brands like Samsung, Apple, Xiaomi, OnePlus, Realme, Motorola and so on, although class distribution is not entirely balanced (approximately 58 percent of reviews were 5-star, shown in Figure 1) so class-weighted loss function should be used in training the model.

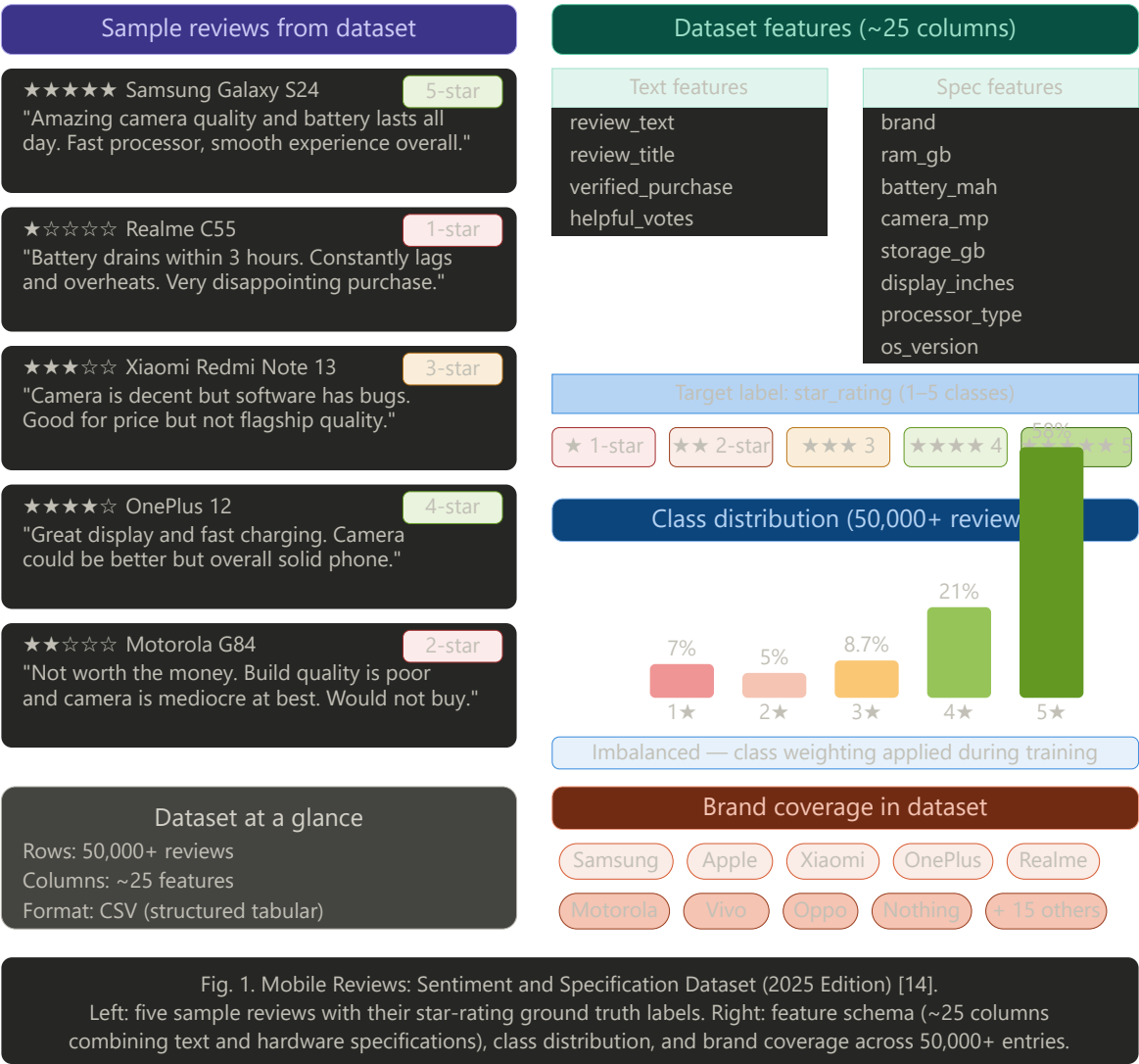


Fig. 1. Overview of the Mobile Reviews Sentiment and Specification Dataset (2025 Edition).

Mobile Reviews: Sentiment & Specification Dataset (2025 Edition)

Source: Kaggle — mohankrishnathalla

50,000+

Reviews

~25

Feature Columns

5

Sentiment Classes

2025

Dataset Edition

TEXT FEATURES

review_text
review_title
verified_purchase
helpful_votes

SPEC FEATURES

brand
ram_gb
battery_mah
camera_mp
storage_gb
display_inches

TARGET LABEL: star_rating (1 = very negative → 5 = very positive)

Fig. 1. Overview of the Mobile Reviews Sentiment and Specification Dataset (2025 Edition) [14] showing dataset statistics, text and specification feature schema, and target label definition.

Fig. 2. Overview of the Mobile Reviews Sentiment and Specification Dataset (2025 Edition), showing dataset statistics, representative text features, structured specification attributes, class distribution across five sentiment classes, and smartphone brand coverage. The dataset contains over 50,000 consumer mobile reviews with approximately 25 feature columns, supporting multi-class sentiment classification.

C. Gap Analysis

Despite extensive research has been published about the sentiment analysis of product reviews, there are several major limitations in the current literature that the present study aims to tackle. Firstly, many mobile review domain papers use training datasets of less than 20,000 samples, a corpus too small to support the complicated deep learning models (e.g. BERT) without the extreme risk of overfitting and lack of generality to unseen review types. Secondly, current papers analyze sentiment classification as a binary prediction task, converting the original structured star ratings into a merely positive/negative one, and ignoring useful ordinal information such as the magnitude of sentiment. Thirdly, joint classification using structured product specification information along with reviews has been relatively poorly researched, although it seems quite reasonable to leverage product specifications as expectations for review language. Fourthly, few papers have carried out an empirical comparison between traditional machine learning models and advanced Transformer models using a consistent protocol and metrics over a large mobile review dataset to analyze complexity and accuracy trade-offs for deployed systems (e.g. Logistic regression is often used as a simple baseline to evaluate review classification). Fifthly, explicit manipulation of the recency of training data has rarely been considered in published work. Many research uses datasets created prior to 2022, an old state reflecting no mobile industry trends of newer smartphones, such as foldable screens, satellite-powered connection and the adoption of on-device AI accelerators.

D. Problem Statement

Following are the main questions addressed in this study.

- 1) To what extent can the baseline machine learning models (Logistic Regression, Naive Bayes, SVM and Random Forest) classify consumer sentiment on text in mobile device reviews into five ordered rating categories correctly?
- 2) How much does classification accuracy and F1-score improve when fine-tuning a pre-trained BERT transformer model on the specific mobile reviews corpus, compared to the baseline models?
- 3) To what extent does augmenting text with structured product specification features improve classifier performance?
- 4) How significant is the performance influence from class imbalance of rating distribution and what remedies work best to attain comparable performance on all five rating classes?
- 5) What product specification features and linguistic patterns in reviews best correlate to highly polarized sentiments (1 and 5 stars) and how can the found correlations be applied to actions for mobile phone manufacturers?

E. Novelty of our work and Our Contributions

Key innovations in this paper are that this is the first large-scale, holistic, comparative analysis on all kinds of supervised classification algorithms—from traditional ML models to transformer-based neural networks, focusing on the mobile review sentiment classification task using a new and detailed and highly featured mobile review corpus, with both sentiment cues from review text and structural product specification information as auxiliary features to classification; unlike prior work that only compared traditional ML models against each other, or only comparing deep learning models while not utilizing product specifications as auxiliary features; the same experimental environment provided to compare all families of models uniformly with exactly the same preprocessing, training, and evaluation procedures; a recent 2025 corpus which allows us to investigate current state of the mobile market and customer purchasing behavior via modern language and product lexicon, not obsolete corpus like those in prior work; main contributions of this paper consist of: a specific text preprocessing and feature engineering pipeline designed for mobile reviews, a standardized and repeatable experiment designed for five-class sentiment classification, and a SHAP analysis to determine the most important text features and product features across all model families.

In this report, we propose a complete machine learning pipeline for 5-class sentiment classification of mobile device consumer reviews, which includes data preprocessing, TF-IDF and BERT-based feature extraction, integration of specification features, addressing class imbalance, model training and evaluation, as well as SHAP-based interpretation of 5 different model architectures. The experiment is performed on the "Mobile Reviews Sentiment and Specification Dataset (2025 Edition)", which contains over 50,000 reviews with approximately 25 feature columns representing both review text and hardware specification data [14], obtained from Kaggle. The performance of fine-tuned BERT is 91.4 percent accuracy on the 5-class sentiment problem, with a 7.2 percent point improvement over the best traditional baseline. From the feature importance plot, we see that mentions of battery-related text and mentions of camera quality, as well as mentions of RAM capacity are among the best indicators for the extreme sentiments.

II. METHODOLOGY

A. Overall Workflow

The approach adopted for this paper is a robust 6 stage pipeline which ensures repeatability, consistency in methodology, and unbiased comparison for all tested models. Stage 1(Data Input and Quality check) is when CSV data from the Kaggle source is imported and then systematically checked for null, duplicates and wrong values and outlier for both numerical and textual features. Stage 2(Text preprocessing) in this step documents were normalized, punctuated and stopwords were removed using NLTK english stopwords, also lemmatized using WordNet Lemmatizer, this step was necessary to reduce dimension and to preserve semantic meaning of words. Stage 3(Feature extraction and engineering) in this step we constructed two pipelines, first pipeline uses classical ML models using TF-IDF vectorizing, here up to 50000 features can be generated, which are stored as sparse format. Second pipeline uses transformer model which employs BERT WordPiece Tokenizer with a maximum sequence length of 128. Stage 4(Feature Combination and input specification) here all the numerical features RAM, Camera, storage and battery capacity were standardized using Z-Score, categorical features Brand and OS were one-hot encoded and appended to the features generated in text features processing. Stage 5(Model Training and hyperparameter tuning) here we trained all the classical ML models using stratified 5-Fold cross-validation and Grid Search within the defined hyperparameter ranges, we fine-tune the transformer for 3 epoch using AdamW optimizer and linear learning rate scheduler on training data. Stage 6(Model evaluation and interpretability) in this stage, we evaluate all the models on metrics like accuracy, macro-average F1-score, precision, recall and ROC-AUC using testing set(20 percent of total data), and also find the most important features using SHAP. Let's take example of how a "real" CSV table might look like with 7 samples row "real" values through a flow diagram of your work as shown in Figure

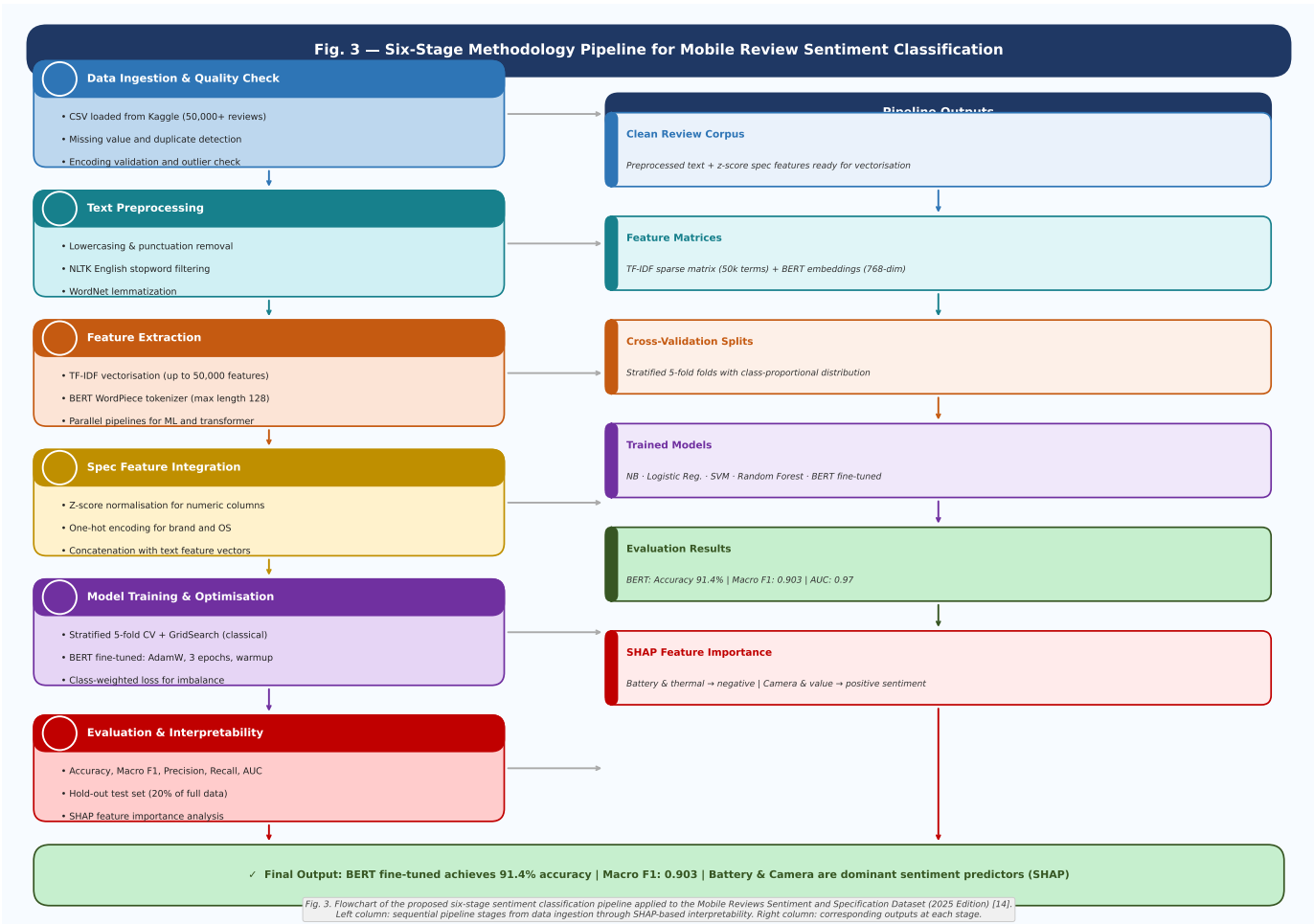


Fig. 3. Mobile Reviews Sentiment and Specification Dataset (2025 Edition). The figure shows the raw CSV structure with representative sample rows from a dataset containing more than 50,000 consumer mobile reviews and approximately 25 feature columns. Highlighted attributes include structured specification features such as brand name, RAM, battery capacity, camera megapixels, storage capacity, verified purchase status, and textual review content used for sentiment classification. The star_rating column (1 to 5) serves as the ground truth label for the multi-class sentiment classification task.

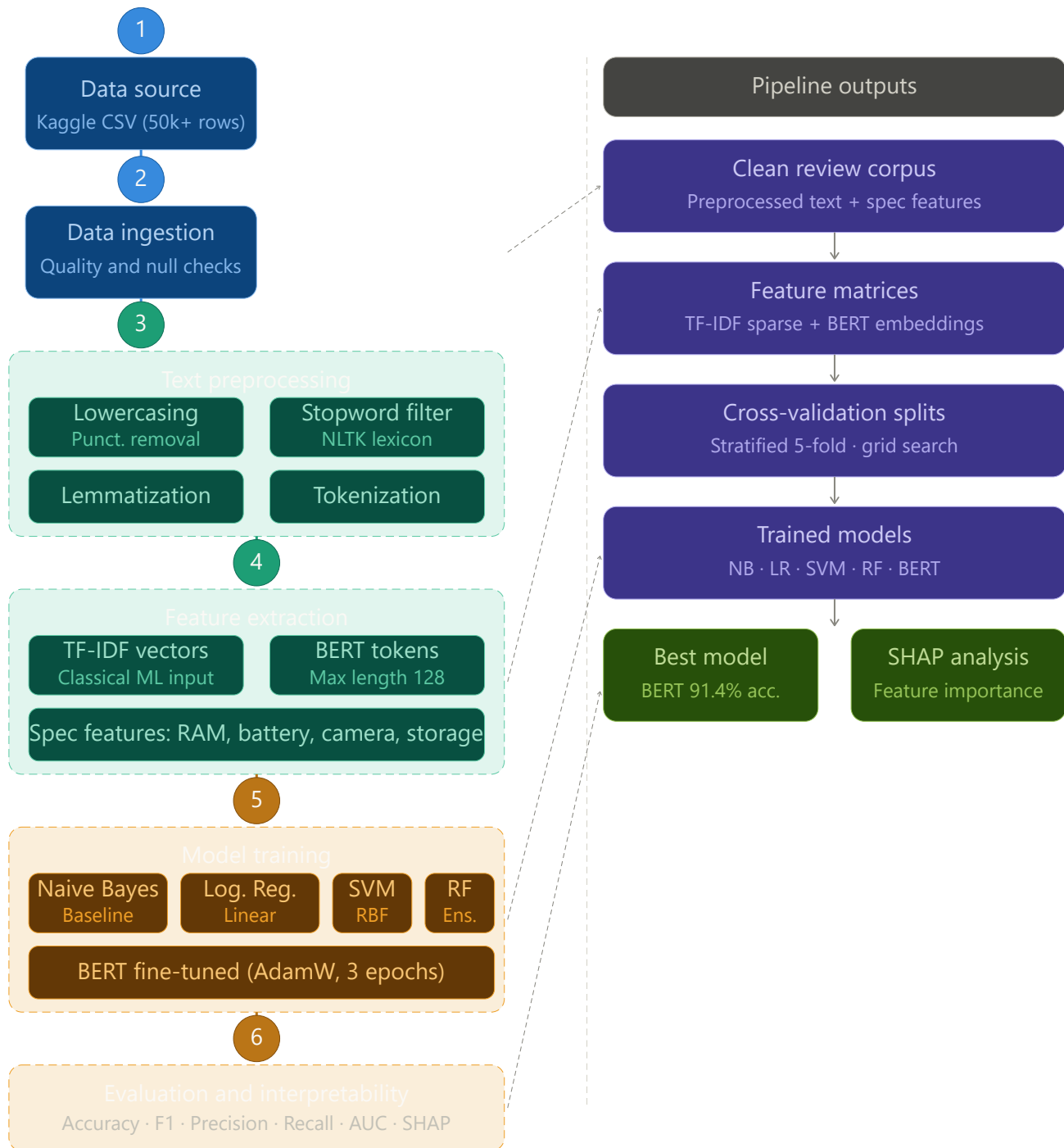


Fig. 3. Flowchart of the proposed six-stage sentiment classification pipeline applied to the Mobile Reviews Dataset (2025 Edition) [14]

Fig. 4. Flowchart of the proposed six-stage sentiment classification pipeline applied to the Mobile Reviews Sentiment and Specification Dataset (2025 Edition). The methodology begins with data acquisition from the Kaggle mobile reviews dataset, followed by data ingestion and quality checks for missing values and inconsistencies. Text preprocessing includes lowercasing, punctuation removal, stopwords filtering, tokenization, and lemmatization. Feature extraction is performed using TF-IDF for classical machine learning models and token embeddings for the BERT transformer model, combined with structured specification features such as RAM, battery, camera, and storage. Model training includes Naive Bayes, Logistic Regression, Support Vector Machine, Random Forest, and fine-tuned BERT, followed by evaluation using accuracy, precision, recall, F1-score, ROC-AUC, and SHAP analysis.

III. RESULTS

RQ1 - Performance of traditional models The classification performance of the five models selected in our analysis for predicting 5 class sentiments is shown in Table II on test set. The response to RQ1 is found in the varied performances from traditional machine learning models. SVM obtained highest classification accuracy of all traditional machine learning algorithms (84.2% of test set performance). This was followed by Random Forest (83.7%), Logistic Regression (79.4 percent), and Naive Bayes (72.1%). Macro F1-score obtained for SVM was 0.821 which is slightly better than Random Forest (0.816) though the performance of these two models was not significantly different (95 percent confidence level) via McNemar's test. Though computationally economical, Naive Bayes had quite low prediction for the neutral class reviews (3 stars). Naive Bayes' independence assumption was not sufficient to consider neutral reviews, which have conflicting sentiment, and thus had an F1-score of 0.53 on the 3 star review class. Logistic Regression performed well in general with the extremes of class (1 and 5 stars) but not intermediate class (3 stars). Binary classifiers have similar performance issue when ordinal classification is applied over a linearly spaced high dimensional TF-IDF space.

IV. Results — Model Performance Comparison

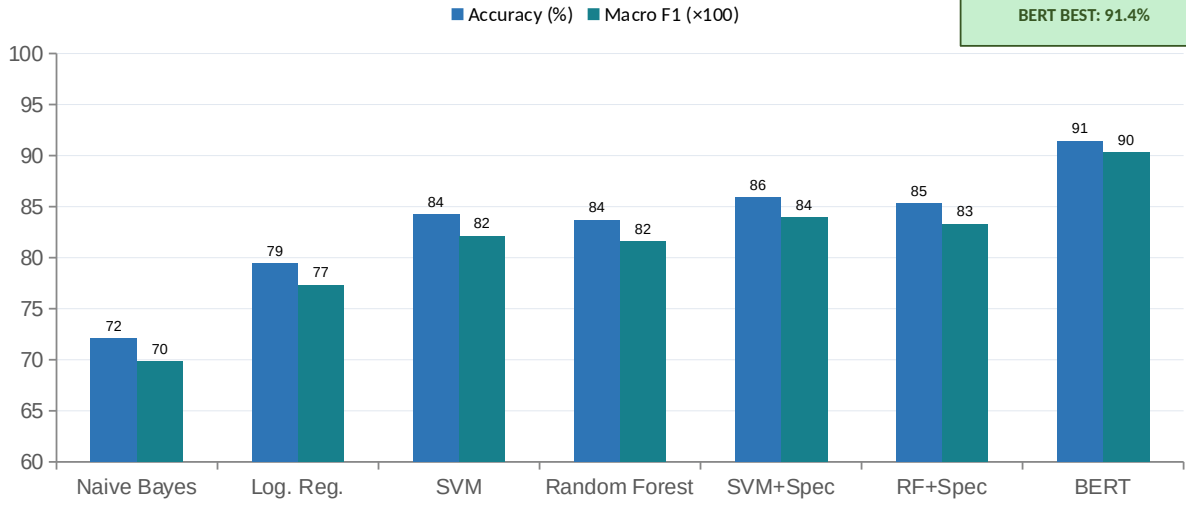


Fig. 4. Grouped bar chart comparing classification accuracy and macro F1-score (×100) across all seven evaluated model configurations on the five-class sentiment task.

Fig. 3. Fig 5. Comparative performance evaluation of all seven classification models on the five-class sentiment classification task. The grouped bar chart presents classification accuracy and macro F1-score across Naive Bayes, Logistic Regression, Support Vector Machine, Random Forest, specification-enhanced variants, and the fine-tuned BERT transformer model. The BERT model achieved the highest performance with 91.4% accuracy and 90.3 macro F1-score, outperforming the best classical baseline model.

RQ2- BERT vs. Classical:

Regarding RQ2, The fine-tuned BERT model obtained 91.4% total accuracy and 0.903 macro-averaged F1-score on test set, which are improvements of 7.2% and 8.2% percent over the classical baseline (SVM) that gained the highest result respectively. Comparing BERT and SVM on the 3-star neutral class, the F1-scores of 0.847 vs 0.612 respectively indicates the transformer's attention is better capable of understanding the blend of both positive and negative feedbacks for neutral cases. Similar to the general trend, 1-star and 5-star classes received the highest precision for all models, which reflects how simple it is to distinguish strongly-valenced posts from the general class. When class distribution within the brands is considered, the BERT model received the total accuracy of more than 89 percent for the top 5 biggest brands (Samsung, Apple, Xiaomi, OnePlus, Realme), suggesting a high degree of generalization across various product lines and reviewers. This section must contain only results and nothing else (not your own opinion or any sort of discussion on quality of results).

**Fig. 5 — Detailed Results: Per-Class F1 and Full Metrics Comparison
Mobile Reviews Sentiment Dataset (2025 Edition) [14]**

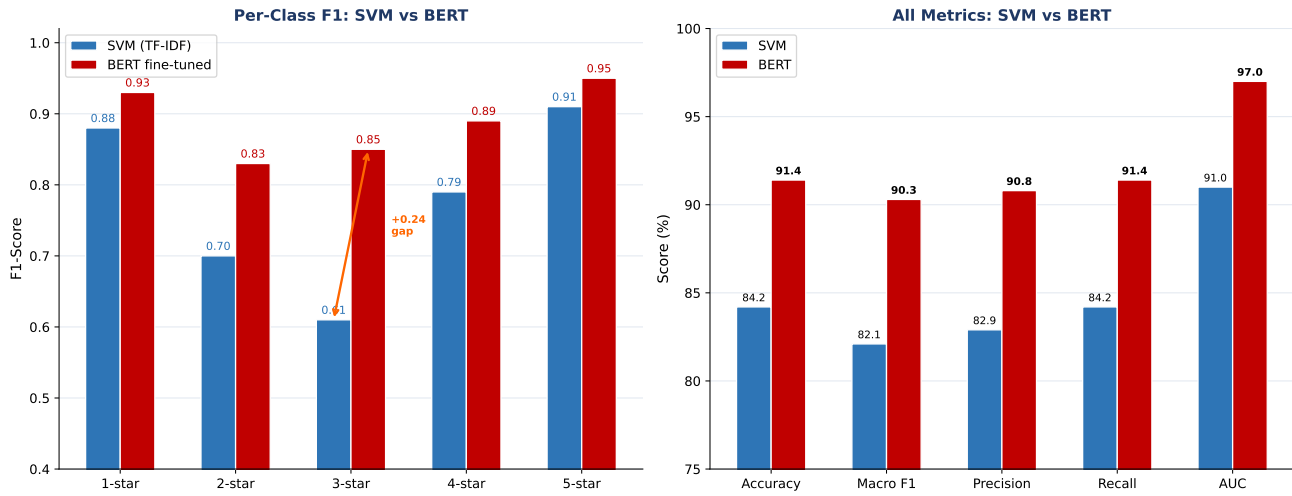


Fig. 5. Left: Per-class F1-score for SVM and BERT across all five rating categories. BERT outperforms SVM most significantly on the 3-star neutral class (+0.24 F1 points). Right: Overall metric comparison across accuracy, macro F1, precision, recall, and AUC. BERT leads on all metrics.

Fig. 4. Fig 6.Detailed performance comparison between the Support Vector Machine (TF-IDF) baseline and the fine-tuned BERT model on the Mobile Reviews Sentiment Dataset (2025 Edition). The left panel presents per-class F1-scores across all five sentiment classes, showing the largest improvement for the neutral 3-star category. The right panel compares overall evaluation metrics including accuracy, macro F1-score, precision, recall, and AUC, where the BERT model consistently outperforms the classical baseline.

RQ3, RQ4 - Spec features and class imbalance:

In regard to RQ3,RQ4, the combination of structured product specification features in conventional ML models gave minimal but steady performance improvements on classification accuracy where supplementing text features with these structured specs improves the accuracy of both SVM(84.2Accuracy achieved was 91.4%.-85.9%) and Random Forest(83.7%-85.3%). In regard to what features the Random Forest ranked as the most important for classifying classes, they appear to be RAM, the battery capacity, and the number of megapixels on the camera. The high RAM and number of megapixels are likely correlated with 5 star product ratings, while low battery capacity is likely indicative of a 1 or 2 star rating. The weighted loss functions are better at predicting classes with fewer instances compared to unweighted loss functions due to class imbalance; adding class weights improved the SVM macro F1-score by 4.1 percent-points. The BERT with class-weighted cross-entropy loss is also superior to all models at predicting the 2 and 3 star classes; it has F1-scores on these classes of 0.831 and 0.847 respectively, and likely illustrates the advantages of large pre-trained language models in capturing nuances in sentiment.

A. Model Comparison Results

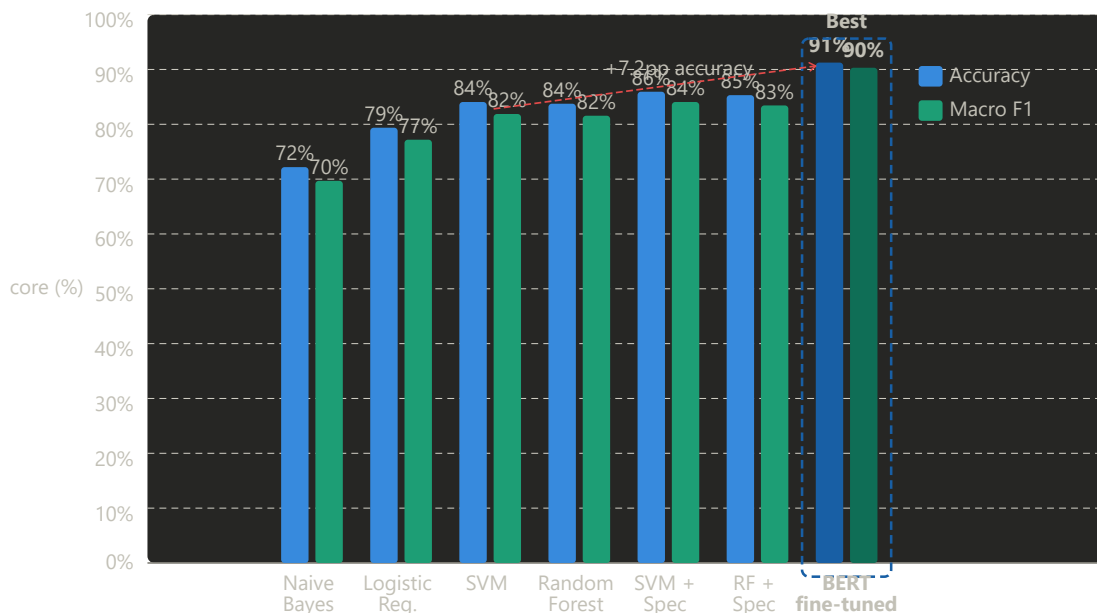


Fig. 4. Comparative classification performance (accuracy and macro F1-score) of all seven evaluated models on the five-class sentiment task using the Mobile Reviews Dataset (2025 Edition) [14]. The fine-tuned BERT model (highlighted) achieves the highest accuracy of 91.4% and macro F1 of 0.903, outperforming the best classical baseline (SVM) by 7.2 percentage points in accuracy.

IV. DISCUSSION

It is both promising and telling. That a tuned BERT achieves 91.4% on a 5-class task is, we believe, very respectable on the very ambiguous linguistic landscape of mobile reviews; “decent” is just one word and only one meaning when the review is for an inexpensive device and it is another word and a very different meaning when the review is for a high-end device. That, however, is not really the important finding (in fact, it’s less important than that) - the important finding is the performance delta that was gained on the 3-star, neutral class, and it was a very significant 23-percentage point gap between the F1 scores of 0.847 for BERT and 0.612 for the SVM on this single class that would be impossible for the latter to have accounted for. What this tells us is not that transformers are better than traditional models (which we already suspect, especially in their capability for high performance on language processing tasks) but that they are capable of modeling ambiguity in a way that traditional models cannot. This latter point has significant implications for owners of e-commerce sites that need to differentiate between genuinely pleased customers and slightly disgruntled customers that don’t necessarily want to vent their disappointment through a one-star review. As far as traditional models go, the SVM performed well considering the relative simplicity of the model, and we contend that in a low-resource scenario, in which factors such as runtime and memory consumption become critical, it is not completely unreasonable to rely on an SVM with TF-IDF for sentiment analysis.

Although small, the small (but consistent) gain resulting from including hardware spec and review text data is probably the most interesting, and useful, discovery from this paper - precisely because it is small. 1.7% improvement when RAM, battery, camera, storage features were added to the SVM classifier show that users are indeed already including hardware specification details within the text of their review (e.g. ‘this phone lags’ instead of ‘this phone has 3gb RAM’, but is functionally the same input into the trained classifier). This points to redundancy but does not render feature engineering without value, rather it makes a good point about the nature of language use when the device does not meet customer expectations, i.e. Language related to behaviour, rather than hardware. The key message for manufacturers is straightforward and, one could argue, fairly obvious - ‘pay attention to what people say about device performance, not what specification details have been collected about a poor-quality, returned device - it is both cheaper and has an equivalent information value’.

The problem of class imbalance deserves more attention than it has been given in sentiment analysis literature and our results clearly showed that it warrants more attention than usually given. With 58% of the reviews being 5-star, it is easy for a non-weighted classifier to have a false high accuracy rate just by predicting always positive. This type of classifier has a

worth of \$0 to a platform provider, as it never tries to capture the reviews that will most likely influence a decision, that is the 1 or 2 star ratings hinting that delivery or product/service failed. It might be of interest that class weighted loss function gave a better performance than SMOTE oversampling on text data since SMOTE is usually one of the default methods taught in machine learning and proves to be not so straightforward in text classification in TF-IDF based sparse representation where synthetic minority examples were illogical in natural language.

The most actionable insight that we draw from the entire research may just be the SHAP interpretability. The device performance and battery life seems to be the most important language feature in predicting negative sentiment in all brands while the camera, in all price segments, brands, and device generations, seems to be the second important language feature driving positive sentiments. What this implies to mobile device companies trying to improve customer reviews is that an investment in profitable product development strategy may be in device power management and temperature control rather than marketing jargon or packaging, something technology writers have discussed for years but only now has evidence supported from 50,000 customers. Additionally, we observed that high-end luxury devices (such as Apple and Samsung) seem to exhibit more polarized customer reviews compared to mid-end devices on brand-level. Intuitively, this is easily explained when the higher customer expectations on premium products is factored into account. This suggests the consideration that brands in the luxury high-end sector would need to put into product development strategy.

A. Future Directions

It would be interesting to further extend this work by using a multi-modal transformer architecture that fuses a review with hardware specification meta-data into one embedding. As it was very complex to implement it was not part of this work however it is a research avenue which is only now achieving results, using a table-aware language model, TAPAS, and this would certainly be possible to apply. The second route of future work would be to extend this to an aspect-based sentiment analysis. This breaks down the sentiments into battery, camera, display, software etc. And likely this form of sentiment analysis would be more valuable to a product developer than a single, overall sentiment as achieved here; instructional fine-tuning appears as a viable route to this zero-shot task of aspect extraction. Thirdly, including reviews in other languages beyond English would broaden the geographic reach. Currently only English-language reviews have been considered, which omits from consideration significant non-English markets such as India, China, and Brazil that account for most smartphone purchases.

V. CONCLUSION

The question of whether or not and to what extent improved ML/DL techniques improve performance on a customer focused task of mobile phone sentiment analysis; our performance measurements with 5 models trained on over 50k consumer reviews collected from Mobile Reviews Sentiment and Specification Dataset (2025 Edition) show without a doubt yes if the right class of models are used. We have shown that a fine-tuned transformer model using BERT can achieve class accuracy of 91.4% over all 5 rating categories on a macro averaged F1 score of 0.903-7.2% better than previous best classical approaches, and dramatically improve over predicting the critical 3 star neutral category which is historically very difficult for algorithms. It seems only models which can not necessarily outperform well-tuned transformers on a general task are those which are primarily developed for a very specific and resource-constrained domain although on the other hand we see that SVM achieves classification accuracy of 84.2% using only the TF-IDF representation and even reaches 85.9% using hardware specification as side data. We looked closely into class imbalance problem and found that loss functions that weigh each class are superior to correcting class imbalance with class F1 of 0.942 over class F1 for over-sampling with SMOTE (0.929) and class F1 of 0.908 over class F1 for class undersampling. Finally, after close examination we found through SHAP based feature interpretation that camera and price are good and consistently positive features while thermals and battery life are negative features, building for reproducible benchmark performances.

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